NETWORKING

Subnets & CIDR Visual Guide

Carving address space —
mask math, subnet design, and VPC layouts.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Mask Math: Bits Decide

The mask splits 32 bits into network part and host part.

```
IP:   192.168.1.130  =  11000000.10101000.00000001.10000010
Mask: 255.255.255.0  =  11111111.11111111.11111111.00000000  (/24)
AND → 192.168.1.0    =  network address (host bits zeroed)
```

Prefix length = number of 1-bits. /24 means the first 24 bits are network.

- CIDR /n is just shorthand for the mask — /24 = 255.255.255.0, /16 = 255.255.0.0.
- Hosts = 2^(32-n) – 2 (subtract network + broadcast). /24 → 254 usable hosts.
- Bitwise AND of IP and mask always gives the network address.

STEP 2/6

Subnetting a Block: Worked Example

Split 192.168.1.0/24 into 4 equal subnets — borrow 2 bits.

192.168.1.0/24 — 256 addresses

borrow 2 host bits → 4 × /26 (64 addresses each)

192.168.1.0/26

.1 – .62 • host .63 bcast

192.168.1.64/26

.65 – .126 • host .127 bcast

192.168.1.128/26

.129 – .190 • host .191 bcast

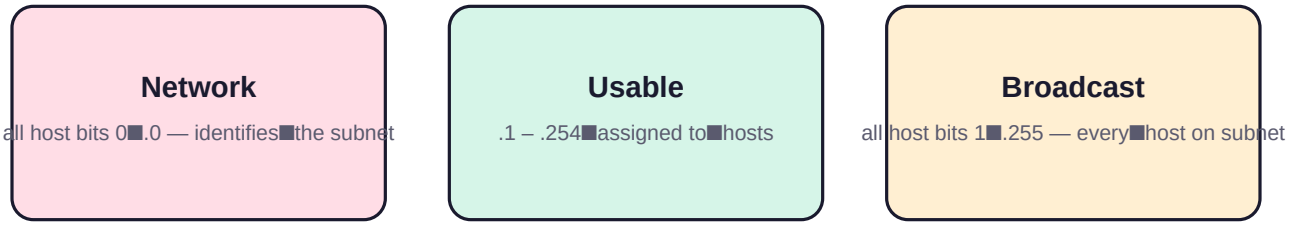
192.168.1.192/26

.193 – .254 • host .255 bcast

STEP 3/6

Network, Broadcast, Usable

Two addresses in every subnet are spoken for.



- Routers use the network address in routing tables; hosts never configure it on an interface.
- Broadcast reaches every host in the subnet — DHCP DISCOVER and ARP rely on it.
- First usable (.1) is the conventional gateway address — convention, not law.

STEP 4/6

The CIDR Cheat Sheet

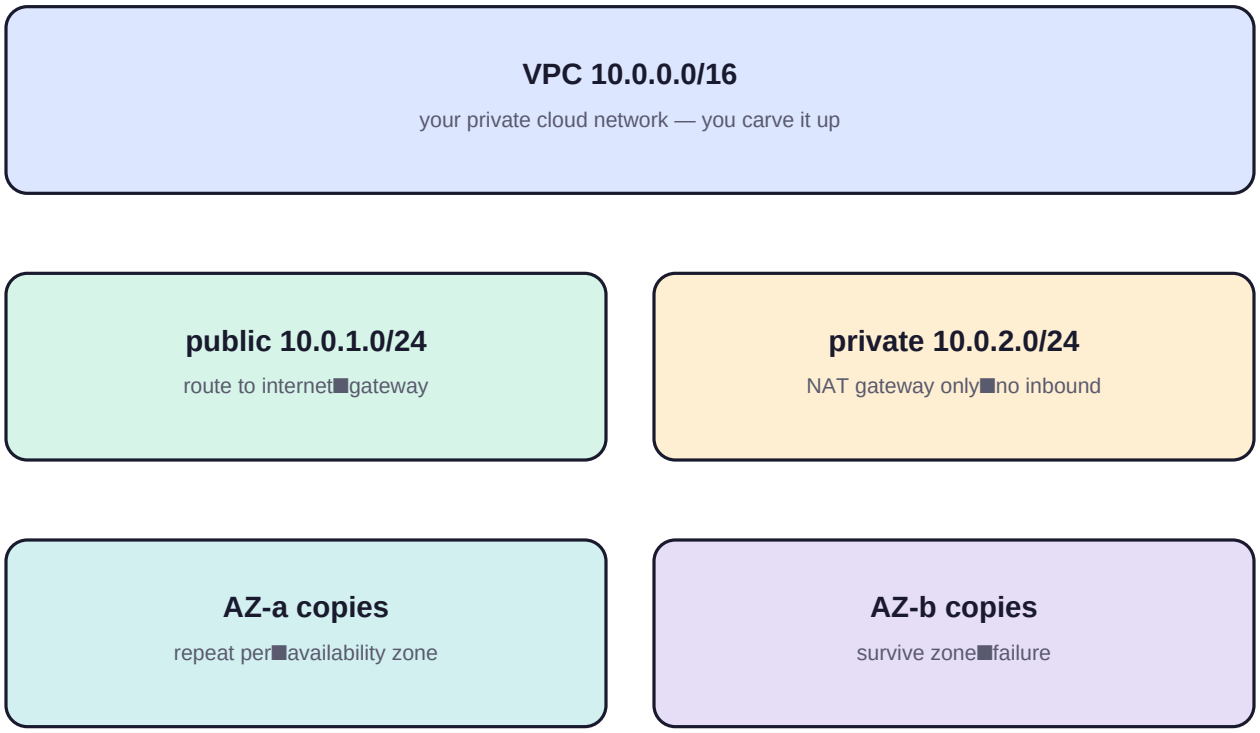
Memorize the common ones; derive the rest.

/32	1 addresses — single host route
/30	4 (2 usable) addresses — point-to-point links
/29	8 (6 usable) addresses — tiny subnets
/28	16 (14 usable) addresses — small VLANs
/24	256 (254 usable) addresses — the classic LAN
/16	65,536 addresses — large org / B-class
/8	16.7M addresses — huge (10.0.0.0/8)

STEP 5/6

VPC Subnet Design

Cloud networks are subnets with extra rules.

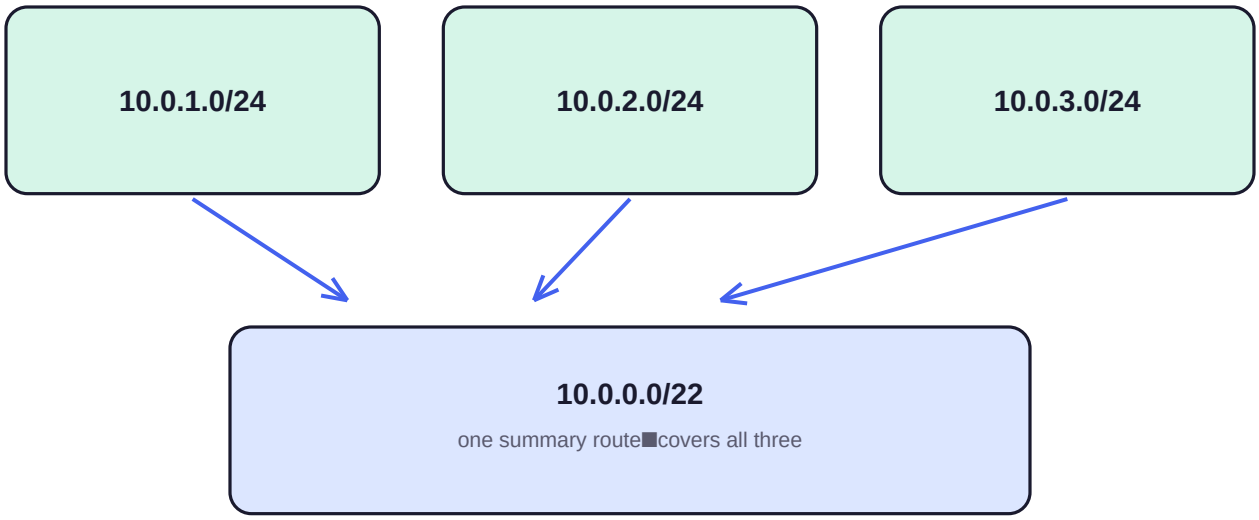


Reserve space: subnets can't be resized later — leave gaps (10.0.10.0/24, 10.0.20.0/24…) for growth.

STEP 6/6

Supernetting & Route Aggregation

CIDR works in reverse too — combine routes to shrink tables.



Internet routers hold ~1M routes only because ISPs aggregate — without it the table would be billions of rows.

STEP ✓/✓

Key Takeaways

Subnets & CIDR — Remember

- The mask splits bits: /n = first n bits are network; hosts = 2^(32–n) – 2.
- Subnet by borrowing host bits: /24 → four /26s of 64 addresses each.
- Network (all-0) and broadcast (all-1) addresses are never assigned to hosts.
- VPC design: public/private subnets per AZ, with gaps left for growth.
- Supernetting aggregates routes — it's why the global routing table stays near ~1M entries.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev